Assignment Linux SS2
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1. Use the find command to search for all .txt files in a directory.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ find . -name "*.txt"
./cdac.txt
./sample.txt
./text_2.txt
./example.txt
./filelist.txt
./text_1.txt
./log.txt
./text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

2. Compare two files using the diff command.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ diff file1 filelist.txt
1,6c1,4
< error
< linux
< Linux
< Linux123
< linux1234
< error123
---
> filelist.txt
> text_1.txt
> text_2.txt
> text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

3. Display the file type of a file using the file command.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ file filelist.txt
filelist.txt: ASCII text
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

4. Create a symbolic link to a file using ln -s.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ln -s sample.txt sample
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls
cdac.txt  example.txt  file1  filelist.txt  log.txt  sample  sample.txt  text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

5. Compress a file using gzip.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ gzip example.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls
cdac.txt  example.txt.gz  file1  filelist.txt  log.txt  sample  sample.txt  text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

6. Decompress a file using gunzip.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ gunzip example.txt.gz
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls
cdac.txt  example.txt  file1  filelist.txt  log.txt  sample  sample.txt  text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

7. Create a zip archive of multiple files.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls
cdac.txt  example.txt  file1  filelist.txt  log.txt  sample  sample.txt  text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ zip archive.zip text_1.txt text_2.txt text_3.txt
  adding: text_1.txt (stored 0%)
  adding: text_2.txt (stored 0%)
  adding: text_3.txt (stored 0%)
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls
archive.zip  cdac.txt  example.txt  file1  filelist.txt  log.txt  sample  sample.txt  text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

8. Extract a zip archive.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ unzip archive.zip
Archive:  archive.zip
replace text_1.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
 extracting: text_1.txt
 extracting: text_2.txt
 extracting: text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls
archive.zip  cdac.txt  example.txt  file1  filelist.txt  log.txt  sample  sample.txt  text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

9. Create a tar archive of a directory.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ tar -cvf archive.tar Folder_Dir
Folder_Dir/
Folder_Dir/cdac.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls
archive.tar  archive.zip  cdac.txt  example.txt  file1  filelist.txt  Folder_Dir  log.txt  sample  sample.txt  text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

10. Extract files from a tar archive.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ tar -xvf archive.tar
Folder_Dir/
Folder_Dir/cdac.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls
archive.tar  archive.zip  cdac.txt  example.txt  file1  filelist.txt  Folder_Dir  log.txt  sample  sample.txt  text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

11. Create a file using the touch command.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ touch Linux_ss2.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls
archive.tar  archive.zip  cdac.txt  example.txt  file1  filelist.txt  Folder_Dir  Linux_ss2.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

12. Display currently logged-in users using who.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ who
digvijay tty2         2026-03-11 21:09 (:0)
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

13. Display the current username using whoami.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ whoami
digvijay
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

14. List running processes using ps.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ps
    PID TTY          TIME CMD
   1680 pts/0    00:00:01 bash
   7074 pts/0    00:00:00 ps
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

15. Kill a running process using kill.

```
digvijay    7035    2503  0 21:44 ?        00:00:00 /snap/firefox/
root        7066       2  0 21:45 ?        00:00:00 [kworker/1:0-c
digvijay    7076    1680  0 21:45 pts/0    00:00:00 ps -ef
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ kill 2503
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ kill 2330
```

16. Create 100 files named file1 to file100 using a single command.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux/Files
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files$ touch file{1..100}.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files$ ls
file100.txt  file17.txt  file24.txt  file31.txt  file39.txt  file46.txt  file53.txt  file60.txt  file68.txt  file75.txt  file82.txt  file8.txt   file97.txt
file10.txt   file18.txt  file25.txt  file32.txt  file3.txt   file47.txt  file54.txt  file61.txt  file69.txt  file76.txt  file83.txt  file90.txt  file98.txt
file11.txt   file19.txt  file26.txt  file33.txt  file40.txt  file48.txt  file55.txt  file62.txt  file6.txt   file77.txt  file84.txt  file91.txt  file99.txt
file12.txt   file1.txt   file27.txt  file34.txt  file41.txt  file49.txt  file56.txt  file63.txt  file70.txt  file78.txt  file85.txt  file92.txt  file9.txt
file13.txt   file20.txt  file28.txt  file35.txt  file42.txt  file4.txt   file57.txt  file64.txt  file71.txt  file79.txt  file86.txt  file93.txt
file14.txt   file21.txt  file29.txt  file36.txt  file43.txt  file50.txt  file58.txt  file65.txt  file72.txt  file7.txt   file87.txt  file94.txt
file15.txt   file22.txt  file2.txt   file37.txt  file44.txt  file51.txt  file59.txt  file66.txt  file73.txt  file80.txt  file88.txt  file95.txt
file16.txt   file23.txt  file30.txt  file38.txt  file45.txt  file52.txt  file5.txt   file67.txt  file74.txt  file81.txt  file89.txt  file96.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files$
```

17. Create a directory dbda and inside it create a directory test.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux/Files/dbda
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files$ mkdir -p dbda/test
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files$ ls
dbda          file16.txt  file23.txt  file30.txt  file38.txt  file45.txt  file
file100.txt   file17.txt  file24.txt  file31.txt  file39.txt  file46.txt  file
file10.txt    file18.txt  file25.txt  file32.txt  file3.txt   file47.txt  file
file11.txt    file19.txt  file26.txt  file33.txt  file40.txt  file48.txt  file
file12.txt    file1.txt   file27.txt  file34.txt  file41.txt  file49.txt  file
file13.txt    file20.txt  file28.txt  file35.txt  file42.txt  file4.txt   file
file14.txt    file21.txt  file29.txt  file36.txt  file43.txt  file50.txt  file
file15.txt    file22.txt  file2.txt   file37.txt  file44.txt  file51.txt  file
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files$ cd dbda/
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda$ ls
test
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda$
```

18. Move all .txt files into the test directory.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux/Files/dbda/test
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files$ mv *.txt dbda/test
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files$ cd dbda/test/
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda/test$ ls
file100.txt  file17.txt  file24.txt  file31.txt  file39.txt  file46.txt  file53.txt  file60.txt  file68.txt  file75.txt  file82.txt  file8.txt   file97.txt
file10.txt   file18.txt  file25.txt  file32.txt  file3.txt   file47.txt  file54.txt  file61.txt  file69.txt  file76.txt  file83.txt  file90.txt  file98.txt
file11.txt   file19.txt  file26.txt  file33.txt  file40.txt  file48.txt  file55.txt  file62.txt  file6.txt   file77.txt  file84.txt  file91.txt  file99.txt
file12.txt   file1.txt   file27.txt  file34.txt  file41.txt  file49.txt  file56.txt  file63.txt  file70.txt  file78.txt  file85.txt  file92.txt  file9.txt
file13.txt   file20.txt  file28.txt  file35.txt  file42.txt  file4.txt   file57.txt  file64.txt  file71.txt  file79.txt  file86.txt  file93.txt
file14.txt   file21.txt  file29.txt  file36.txt  file43.txt  file50.txt  file58.txt  file65.txt  file72.txt  file7.txt   file87.txt  file94.txt
file15.txt   file22.txt  file2.txt   file37.txt  file44.txt  file51.txt  file59.txt  file66.txt  file73.txt  file80.txt  file88.txt  file95.txt
file16.txt   file23.txt  file30.txt  file38.txt  file45.txt  file52.txt  file5.txt   file67.txt  file74.txt  file81.txt  file89.txt  file96.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda/test$
```

19. Count the number of files in a directory using command-line tools.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux/Files/dbda/test
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda/test$ ls | wc -l
100
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda/test$
```

20. Use history to display previously executed commands.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux/Files/dbda/test
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda/test$ history
    1  clear
    2  sudo apt-get update
    3  sudo apt install virtualbox-guest-x11 -y
    4  clear
    5  ls
    6  cd bda/
```

21. Execute the last command again using !!.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux/Files/dbda/test
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda/test$ ls | wc
    100     100    1092
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda/test$ !!
ls | wc
    100     100    1092
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux/Files/dbda/test$
```

22. Change permissions of a file using chmod.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls -l file1
-rw-rw-r--+ 1 digvijay digvijay 46 Mar  6 23:43 file1
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ sudo chmod 777 file1
[sudo] password for digvijay:
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls -l file1
-rwxrwxrwx+ 1 digvijay digvijay 46 Mar  6 23:43 file1
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

23. Give read, write, and execute permissions to the owner of a file.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls -l text_1.txt
-rw-rw-r-- 1 digvijay digvijay 0 Mar  6 23:23 text_1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ chmod u+rwx text_1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls -l text_1.txt
-rwxrw-r-- 1 digvijay digvijay 0 Mar  6 23:23 text_1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

24. Remove all permissions for others on a file.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls -l text_1.txt
-rwxrw-r-- 1 digvijay digvijay 0 Mar  6 23:23 text_1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ chmod o-rwx text_1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls -l text_1.txt
-rwxrw---- 1 digvijay digvijay 0 Mar  6 23:23 text_1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

25. Change the owner of a file using chown.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss2_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls -l text_1.txt
-rwxrw---- 1 digvijay digvijay 0 Mar  6 23:23 text_1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ sudo chown user1 text_1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$ ls -l text_1.txt
-rwxrw---- 1 user1 digvijay 0 Mar  6 23:23 text_1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss2_linux$
```

Assignment Linux SS3
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1. Change the group of a file using chgrp.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ls -l text1.txt
-rw-rw-r-- 1 digvijay digvijay 0 Mar 11 22:32 text1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ sudo chgrp user1 text1.txt
[sudo] password for digvijay:
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ls -l text1.txt
-rw-rw-r-- 1 digvijay user1 0 Mar 11 22:32 text1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

2. Create a file and assign permissions using numeric mode (e.g., 755).

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ touch sample.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ls -l sample.txt
-rw-rw-r-- 1 digvijay digvijay 0 Mar 11 22:35 sample.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ chmod 755 sample.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ls -l sample.txt
-rwxr-xr-x 1 digvijay digvijay 0 Mar 11 22:35 sample.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

3. View the Access Control List of a file using getfacl.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ getfacl sample.txt
# file: sample.txt
# owner: digvijay
# group: digvijay
user::rwx
group::r-x
other::r-x

digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

4. Grant read and write permission to a specific user using setfacl.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ setfacl -m u:user1:rw sample.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ getfacl sample.txt
# file: sample.txt
# owner: digvijay
# group: digvijay
user::rwx
user:user1:rw-
group::r-x
mask::rwx
other::r-x

digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

5. Remove ACL permissions for a user using setfacl.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ setfacl -x u:user1 sample.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ getfacl sample.txt
# file: sample.txt
# owner: digvijay
# group: digvijay
user::rwx
group::r-x
mask::r-x
other::r-x

digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

6. Create a file /tmp/acl_file.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ touch /tmp/acl_file
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ls -l /tmp/acl_file
-rw-rw-r-- 1 digvijay digvijay 0 Mar 11 22:43 /tmp/acl_file
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

7. Configure ACL so that a specific user has full access to the file.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ setfacl -m u:user1:rw /tmp/acl_file
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ getfacl /tmp/acl_file
getfacl: Removing leading '/' from absolute path names
# file: tmp/acl_file
# owner: digvijay
# group: digvijay
user::rw-
user:user1:rw-
group::rw-
mask::rw-
other::r--

digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

8. Mount a filesystem to a directory using the mount command.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ mount /dev/sdb1 /mnt/mydisk
mount: /mnt/mydisk: mount point does not exist.
       dmesg(1) may have more information after failed mount system call.
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

9. Unmount a mounted filesystem using umount.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ sudo umount /mnt/mydisk
[sudo] password for digvijay:
umount: /mnt/mydisk: no mount point specified.
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

10. Open a file using the vi editor.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ vi text1.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ cat text1.txt
This is CDAC ACTS.
Course name is PGCP-BDA.
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

11. Insert text in vi using insert mode.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
This is C-DAC ACTS.
Course name is PGCP-BDA.
~
~
~
-- INSERT --
```

12. Save and exit from vi.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
This is CDAC ACTS.
Course name is PGCP-BDA.
~
~
~
~
:wq
```

13. Delete a line in vi.

To delete line in vi - open file using vi - use command by placing cursor in front of line – dd

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
This is CDAC ACTS.
~
~
~
~
~
                                                    1,1           All
```

14. Search for a word in vi.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
This is CDAC ACTS.
Course name is PGCP-BDA.
~
~
~
~
:s/CDAC/C-DAC/g
```

15. Replace a word in vi.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
This is CDAC ACTS.
Course name is PGCP-BDA.
~
~
~
~
:s/CDAC/C-DAC/g
```

16. Copy and paste text inside vi.
To copy line use yy
To paste line use p

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
This is C-DAC ACTS.
Course name is PGCP-BDA.
Course name is PGCP-BDA.
~
~
~
                                                    3,1           All
```

17. Create a simple Bash script that prints "Hello Linux".

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
  GNU nano 7.2
#!/bin/bash
echo "Hello Linux"
```

18. Make the script executable and run it.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ cat script_1.sh
#!/bin/bash
echo "Hello Linux"
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ls -l script_1.sh
-rw-rw-r-- 1 digvijay digvijay 31 Mar 11 23:11 script_1.sh
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ chmod u+x script_1.sh
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ls -l script_1.sh
-rwxrw-r-- 1 digvijay digvijay 31 Mar 11 23:11 script_1.sh
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ./script_1.sh
Hello Linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

19. Create a Bash script that adds two numbers.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ cat script_2.sh
#!/bin/bash
read -p "Enter first number:" a

read -p "Enter second number:" b

sum=$((a + b))
echo "Sum = $sum"
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ./script_2.sh
Enter first number:10
Enter second number:5
Sum = 15
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

20. Write a script that reads user input using read.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ cat script_2.sh
#!/bin/bash
read -p "Enter first number:" a

read -p "Enter second number:" b

sum=$((a + b))
echo "Sum = $sum"
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ./script_2.sh
Enter first number:10
Enter second number:5
Sum = 15
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

21. Write a script using an if condition.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ cat script_3.sh
#!/bin/bash
read -p "Enter a number:" num

if (( $num%2==0 ))
then
        echo "Number is even."
else
        echo "Number is odd."
fi
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ./script_3.sh
Enter a number:5
Number is odd.
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

22. Write a script that checks whether a file exists.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ cat script_4.sh
#!/bin/bash
echo "Enter filename:"
read file

if [ -f $file ]
then
echo "File exists"
else
echo "File does not exist"
fi
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ./script_4.sh
Enter filename:
text1.txt
File exists
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

23. Write a script using if-else.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ cat script_3.sh
#!/bin/bash
read -p "Enter a number:" num

if (( $num%2==0 ))
then
        echo "Number is even."
else
        echo "Number is odd."
fi
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ./script_3.sh
Enter a number:5
Number is odd.
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

24. Write a script using a case statement.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ cat script_5.sh
#!/bin/bash
echo "Enter a number (1-3):"
read num

case $num in
1) echo "One";;
2) echo "Two";;
3) echo "Three";;
*) echo "Invalid number";;
esac
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ./script_5.sh
Enter a number (1-3):
3
Three
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

25. Write a script that prints numbers from 1 to 10 using a for loop.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss3_linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ cat script_6.sh
#!/bin/bash
echo "For Loop"
for ((i=1;i<=10;i++))
do
        echo $i
done
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$ ./script_6.sh
For Loop
1
2
3
4
5
6
7
8
9
10
digvijay@digvijay-virtualbox:~/Linux_SS/ss3_linux$
```

Assignment Linux SS4
Name: Digvijay Vikas Sawant
PRN: 260240125023

1. Write a script using a while loop.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ cat script.sh
#!/bin/bash
count=1
while [ $count -le 5 ]
do
  echo "Iteration number: $count"
  ((count++))
done
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ ./script.sh
Iteration number: 1
Iteration number: 2
Iteration number: 3
Iteration number: 4
Iteration number: 5
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$
```

2. Write a script that uses command-line arguments.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ cat script.sh
#!/bin/bash
echo "First argument: $1"
echo "Second argument: $2"
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ ./script.sh Hello DS
First argument: Hello
Second argument: DS
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$
```

3. Write a script that displays environment variables.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ cat script.sh
#!/bin/bash
echo "HOME: $HOME"
echo "USER: $USER"
echo "PATH: $PATH"
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ ./script.sh
HOME: /home/digvijay
USER: digvijay
PATH: /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/loc
al/games:/snap/bin
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$
```

4. Write a script that uses positional parameters.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ cat script.sh
#!/bin/bash
echo "Script name: $0"
echo "First parameter: $1"
echo "Second parameter: $2"
echo "Total parameters: $#"
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ ./script.sh hello ds
Script name: ./script.sh
First parameter: hello
Second parameter: ds
Total parameters: 2
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$
```

5. Display all environment variables using a command.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ env
SHELL=/bin/bash
WINDOWID=0
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-Lubuntu:/etc/xdg:/etc:/usr/share
XDG_SESSION_PATH=/org/freedesktop/DisplayManager/Session0
XDG_MENU_PREFIX=lxqt-
SAL_VCL_QT5_USE_CAIRO=true
LANGUAGE=
LC_ADDRESS=en_IN
LC_NAME=en_IN
SSH_AUTH_SOCK=/tmp/ssh-cOZaZV8cyLYa/agent.1149
XDG_DATA_HOME=/home/digvijay/.local/share
XDG_CONFIG_HOME=/home/digvijay/.config
```

6. Create a Git repository locally using git init.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux/my-project
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux$ mkdir my-project && cd my-project
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/digvijay/Linux_SS/ss4_Linux/my-project/.git/
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$
```

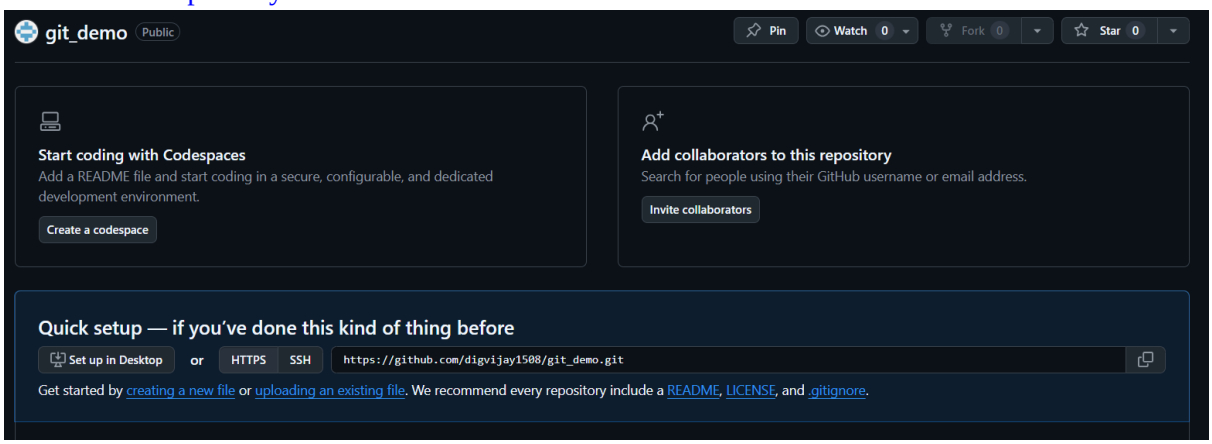
7. Add a file to Git using git add.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux/my-project
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ touch file.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git add file.txt
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file.txt

digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$
```

8. Commit changes using git commit.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux/my-project
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git commit -m "Inatial commit"
[master (root-commit) 3639fc1] Inatial commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file.txt
```

9. Create a repository on GitHub.



10. Clone a GitHub repository using git clone.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux/my-project
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git clone https://github.com/digvijay1508/git_demo.git
Cloning into 'git_demo'...
warning: You appear to have cloned an empty repository.
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$
```

11. Push changes to GitHub using git push.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux/my-project
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git branch -M main
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git remote add origin https://github.com/digvijay1508/git_demo.git
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git push -u origin main
Username for 'https://github.com': digvijay1508
Password for 'https://digvijay1508@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 219 bytes | 219.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/digvijay1508/git_demo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$
```

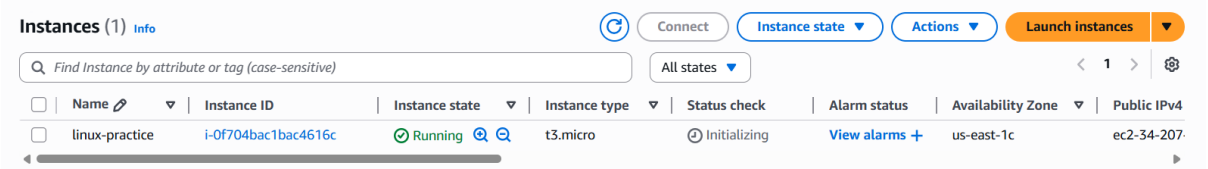
12. Create a branch in Git.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux/my-project
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git branch new_branch
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git checkout new_branch
Switched to branch 'new_branch'
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$
```

13. Merge two branches in Git.

```
digvijay@digvijay-virtualbox: ~/Linux_SS/ss4_Linux/my-project
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git checkout main
Switched to branch 'main'
Your branch is based on 'origin/main', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$ git merge new_branch
Already up to date.
digvijay@digvijay-virtualbox:~/Linux_SS/ss4_Linux/my-project$
```

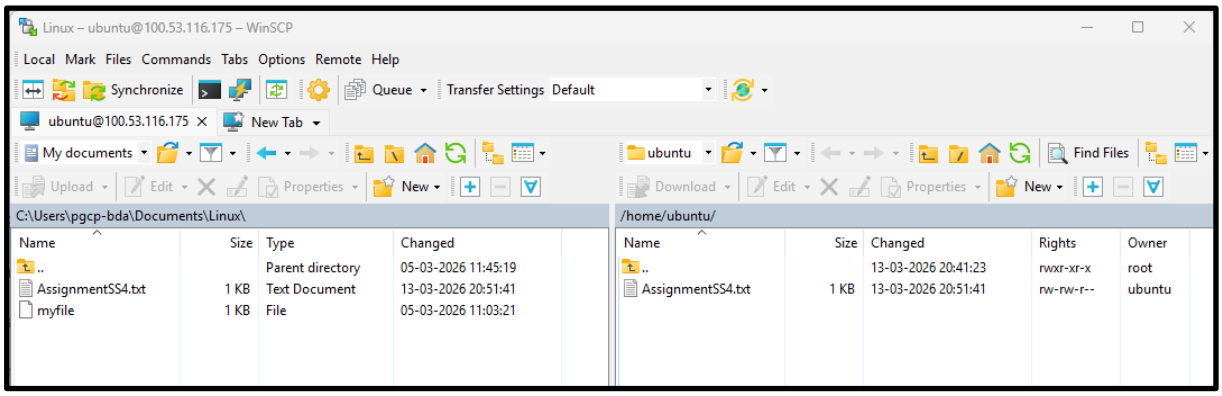
14. Launch a Linux virtual machine in a cloud platform.



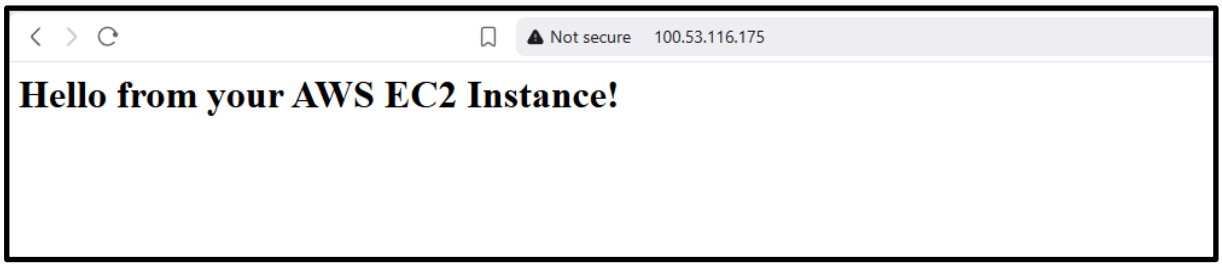
15. Connect to the cloud VM using SSH.

```
ubuntu@ip-172-31-20-21: ~
Using username "ubuntu".
Authenticating with public key "cdac-bda"
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Mar 13 15:13:32 UTC 2026

  System load:  0.18               Temperature:           -273.1 C
  Usage of /:   30.0% of 6.71GB    Processes:             119
  Memory usage: 26%                Users logged in:       0
  Swap usage:   0%                 IPv4 address for ens5: 172.31.20.21

Expanded Security Maintenance for Applications is not enabled.

117 updates can be applied immediately.
76 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-20-21:~$
```

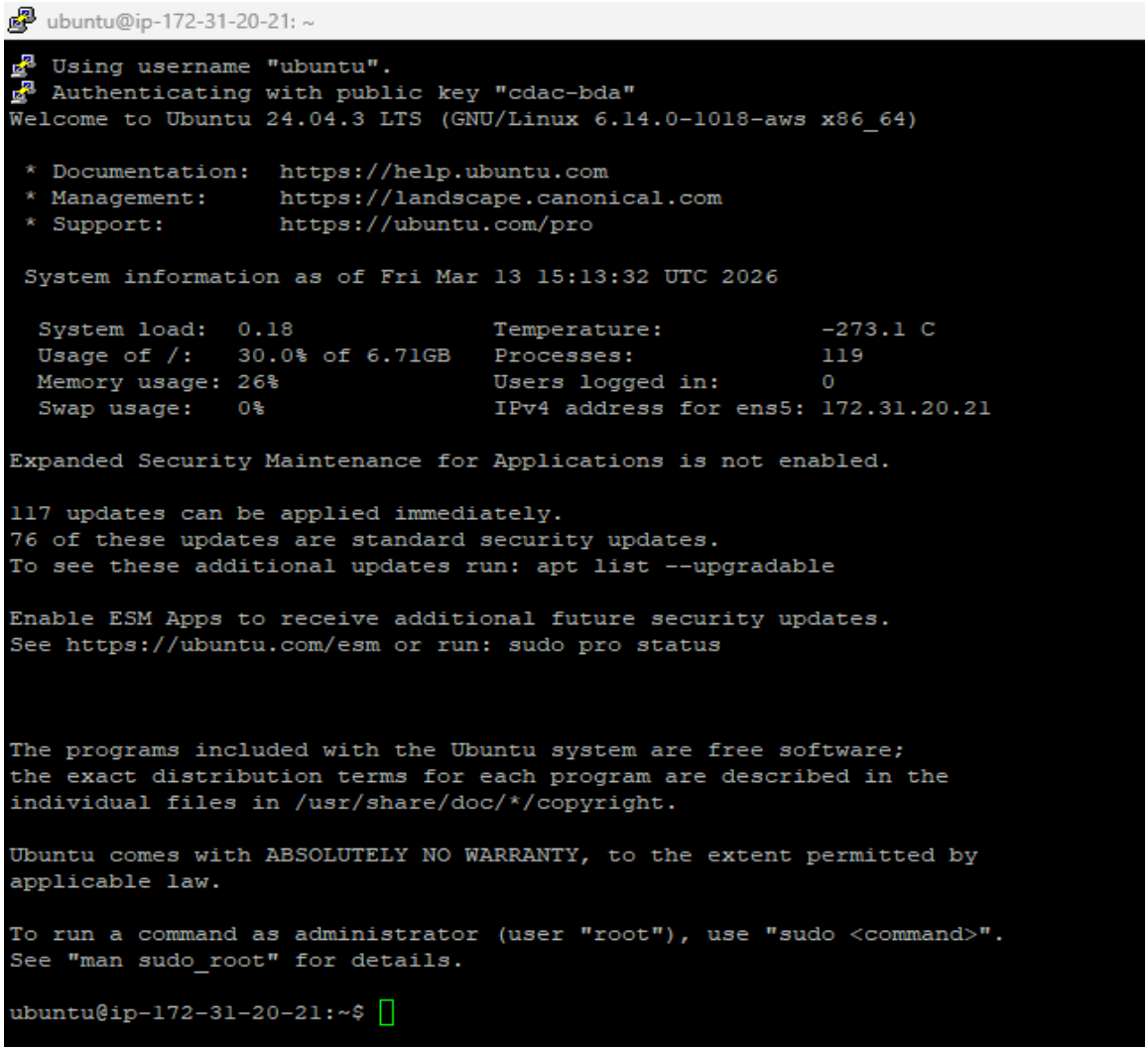
16. Upload a file to the VM using SCP or SFTP.



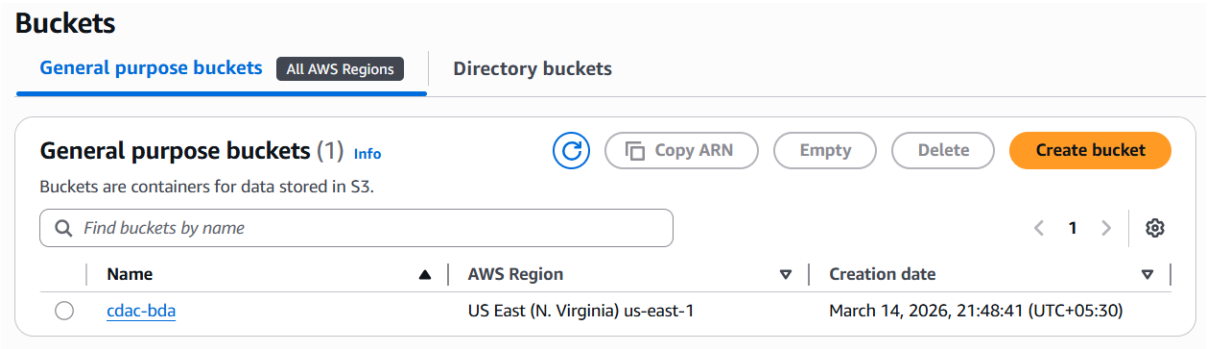
17. Create an AWS EC2 instance.



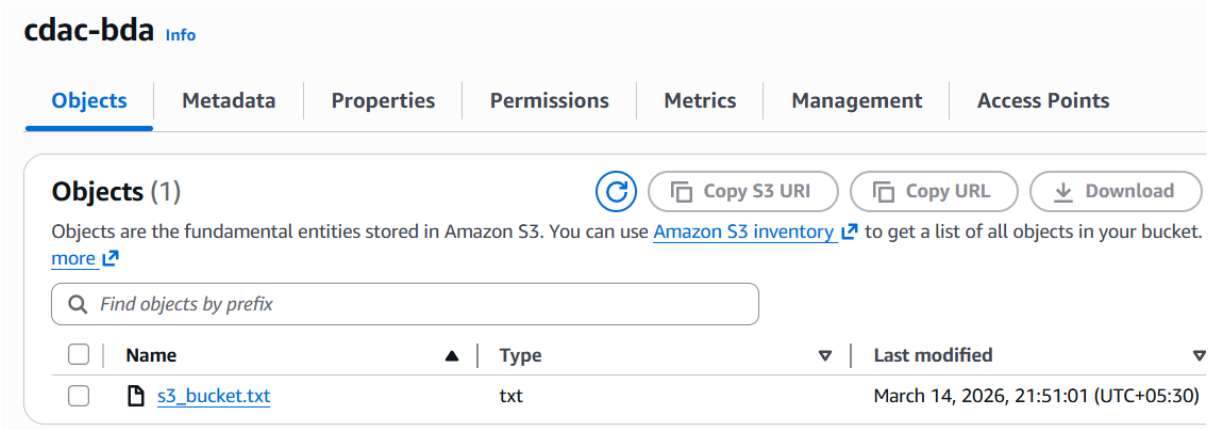
18. Connect to the EC2 instance using SSH.



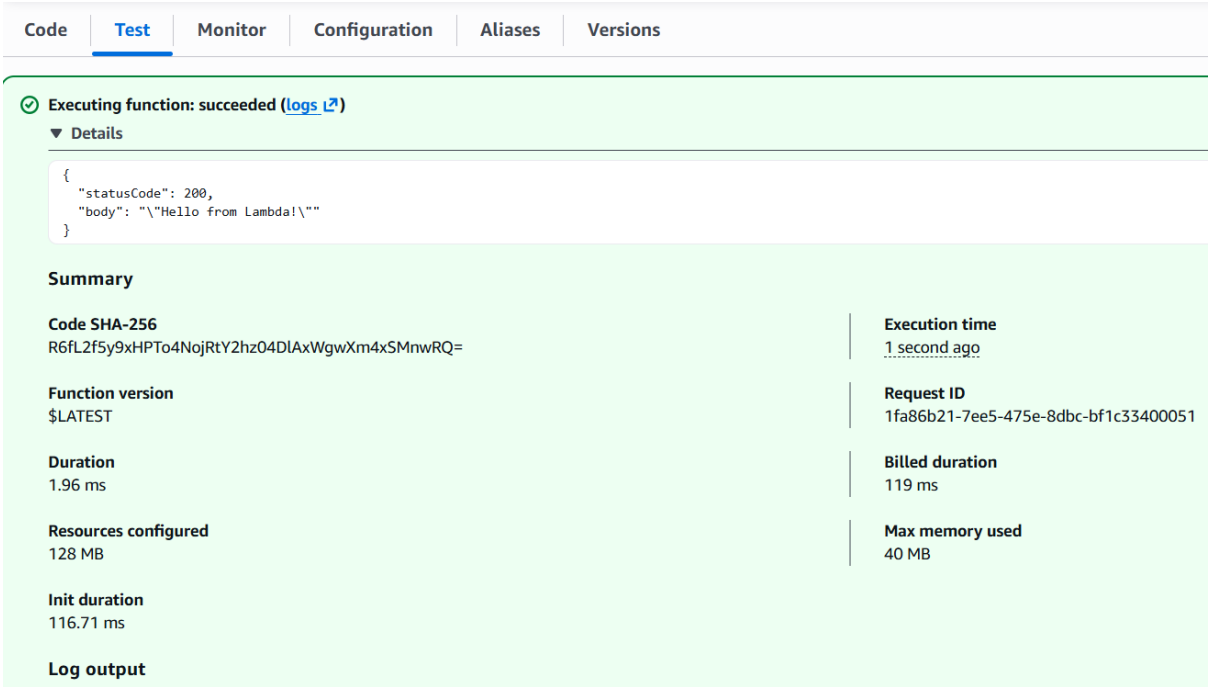
19. Create an AWS S3 bucket.



20. Upload a file to the S3 bucket.



21. Create a simple AWS Lambda function.



22. Launch a Windows Server VM on a cloud platform.

Instance summary for i-012341db6e2e50f2c Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID i-012341db6e2e50f2c	**Public IPv4 address** 13.220.82.190 | open address	**Private IPv4 addresses** 172.31.27.75
IPv6 address –	**Instance state** Running	**Public DNS** ec2-13-220-82-190.compute-1.amazonaws.com | open address
Hostname type IP name: ip-172-31-27-75.ec2.internal	**Private IP DNS name (IPv4 only)** ip-172-31-27-75.ec2.internal	
Answer private resource DNS name IPv4 (A)	**Instance type** t3.micro	**Elastic IP addresses** –
Auto-assigned IP address 13.220.82.190 [Public IP]	**VPC ID** vpc-0b3f8d7a6bce234d9	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role –	**Subnet ID** subnet-0cb73eaed4e096f24	**Auto Scaling Group name** –
IMDSv2 Optional EC2 recommends setting IMDSv2 to required | Learn more	**Instance ARN** arn:aws:ec2:us-east-1:481048082453:instance/i-012341db6e2e50f2c	**Managed** false
Operator –		

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance details Info

AMI ID ami-0f7a0c94dce9ab456	**Monitoring** disabled	**Platform details** Windows
AMI name Windows_Server-2022-English-Full-Base-2026.02.11	**Allowed image** -	**Termination protection** Disabled
Stop protection Disabled	**Launch time** Sat Mar 14 2026 22:21:05 GMT+0530 (India Standard Time) (2 minutes)	**AMI location** amazon/Windows_Server-2022-English-Full-Base-2026.02.11
Instance reboot migration Default (On)	**Instance auto-recovery** Default	**Lifecycle** normal
Stop-hibernate behavior Disabled	**AMI Launch index** 0	**Key pair assigned at launch** –
State transition reason –	**Credit specification** unlimited	**Kernel ID** –
State transition message –	**Usage operation** RunInstances:0002	**RAM disk ID** –
Owner 481048082453	**Enclaves Support** Disabled	**Boot mode** –
Current instance boot mode legacy-bios	**Allow tags in instance metadata** Disabled	**Use RBN as guest OS hostname** Disabled
Answer RBN DNS hostname IPv4		

23. Create a Virtual Private Cloud (VPC).

You successfully created vpc-001bbaa807dc68f53 / my-vpc-01

vpc-001bbaa807dc68f53 / my-vpc-01 Actions

Details Info

VPC ID vpc-001bbaa807dc68f53	**State** Available	**Block Public Access** Off	**DNS hostnames** Disabled
DNS resolution Enabled	**Tenancy** default	**DHCP option set** dopt-0c8b1a6bc90a1e61c	**Main route table** rtb-01ea59a7643b9e084
Main network ACL acl-0196379571b81a53e	**Default VPC** No	**IPv4 CIDR** 10.0.0.0/16	**IPv6 pool** –
IPv6 CIDR (Network border group) –	**Network Address Usage metrics** Disabled	**Route 53 Resolver DNS Firewall rule groups** –	**Owner ID** 481048082453
Encryption control ID –	**Encryption control mode** –		

24. Deploy a simple application to a cloud VM using GitHub.

An application can be deployed to a cloud virtual machine by cloning a repository from **GitHub** and running the application on a cloud instance such as **Amazon EC2**.

Steps

1. Launch a Linux virtual machine using **Amazon EC2**.
2. Connect to the VM using SSH.
   Example command:
   ssh -i keypair.pem ubuntu@public-ip
3. Install Git on the VM:
   Sudo apt update
   sudo apt install git -y
4. Clone the application repository from GitHub:
   git clone https://github.com/username/project.git
5. Move into the project directory:
   cd project
6. Run the application.
   Example (Python application):
   python3 app.py

Result

- The application is downloaded from GitHub.
- The program runs on the cloud VM.

25. Compare the services EC2, S3, and Lambda by listing their purpose and use cases.

Service	Purpose	Type of Service	Common Use Cases
Amazon EC2	Provides virtual servers in the cloud	Infrastructure as a Service (IaaS)	Hosting websites, running applications, machine learning workloads
Amazon S3	Provides scalable object storage	Storage Service	Storing files, backups, images, videos, static website hosting
AWS Lambda	Runs code without managing servers	Serverless computing	Event-driven processing, automation tasks, backend APIs